AFRINEXT · PHASE 2 · FINANCIAL REVIEW GATE

Late Payment Policy

A verified payment can arrive after the checkout expired. The money moved whatever our timer did, so it is never ignored: the order ends as **paid** when fulfilling is still safe, and as **refund_due** when it is not. No confirmed payment is invisible any more.

Branch **claude/create-app-repository-1cxsih** | Code state **4342c06** | Re-verified at **9890cdf**

Tests **349 passed · 24 browser** | Mutations **8 of 8 caught**

PROVENANCE — WHAT THIS DOCUMENT DESCRIBES, EXACTLY

Branch: `claude/create-app-repository-1cxsih`. Not merged into `main`.

Code state under review: `4342c06` — the last commit that touches implementation. Everything after it on this branch is documentation, and that is checkable in one command:

```
git diff 4342c06 HEAD -- ':!docs'   # returns nothing
```

Re-verified at `9890cdf`, the branch head when this revision was prepared: full local suite green, CI run #24 green.

COMMIT	WHAT IT IS	TOUCHES CODE	CI
5d92c8e	The policy: state machine, grace window, row lock, commerce reconciliation, migration 0011, 31 tests, architecture doc	yes	#22 RED browser race, §7
4342c06	The fix for that race. **The last commit touching code.**	yes	#23 GREEN
9890cdf	This review packet, first issue	no	#24 GREEN
this commit	This packet, reissued to correct its own provenance	no	runs on push

WHY THE HEAD SHOWN IS THE CODE STATE AND NOT THE TIP

A document cannot contain the hash of the commit that adds it — writing the hash changes the content, which changes the hash. Any packet that names the branch tip is therefore stale by exactly one commit the moment it is committed, and the previous issue of this packet was stale for precisely that reason.

So the identifier this packet leads with is the one that *stops moving*: the code state. It has not changed since `4342c06`, the command above proves it in one line, and every commit after it is listed in the table with what it contains. Nothing about the implementation under review is ambiguous.

1 What happened before, traced rather than remembered

The investigation ran the real domain code against a real database. This is what it printed:

```
after initiation   : order=pending_payment payment=pending   entitlements=0 fee_schedules=0
after expiry       : order=expired         payment=pending   entitlements=0 fee_schedules=0
webhook outcome    : {"outcome":"applied", "orderStatus":"expired"}
after late confirm : order=expired         payment=succeeded  entitlements=0 fee_schedules=0
reconciliation clean : true
ledger entries       : 0
```

THREE FINDINGS, NONE OF THEM COMFORTABLE

The money left no commercial trace. The payment moved because every check passed. The order's guarded `UPDATE` named `pending_payment`, matched nothing, and the entitlement and fee-schedule writes sat inside that branch.

Nothing surfaced it. Reconciliation compared ledger balances and reported clean, because it knew nothing about orders and the ledger was empty by design. The only trace was one audit row.

The buyer could be charged twice. Repeat purchase is refused by checking `entitlements`. A late payment created none, so the second checkout succeeded — reachable, undetected, and the buyer's only remedy was a support ticket.

The blueprint could not settle it: its order machine is `created → awaiting_payment → paid → fulfilled → settled` and **has no `expired` state at all.** The checkout TTL arrived later. So the question went to review, and came back as option D.

2 The policy, as decided

WHEN THE PAYMENT IS VERIFIED	CONDITIONS	ORDER BECOMES
While the order is `pending_payment`	the ordinary path, unchanged	`PAID` entitlement + frozen fees
After expiry, **at or before** expiry + 24 h	product published · store published · buyer does not already own it	`PAID` + `late_payment_at`
After expiry, **more than** 24 h later	—	`REFUND_DUE`
After expiry, inside 24 h, any condition failing	—	`REFUND_DUE`
A **failed** payment after expiry	no money arrived	`STAYS EXPIRED`

> **WHAT REFUND_DUE MEANS, IN THE WORDS USED EVERYWHERE**
>
> "Payment successfully confirmed, but Afrinext did not fulfil the order and must later execute a refund through the appropriate payment provider."
>
> It is a **queue and a state.** It is not an executed refund, not a request sent to anybody, and not a credit. **No refund execution exists in the codebase**, and nothing drains this queue. The buyer's screen says exactly that, in French and English, and stops there.

The boundary is inclusive, and tested at the millisecond

Grace applies when the payment is verified **at or before** `expires_at + 24 h`. One millisecond later is `refund_due`. That is why `applyProviderEvent` now takes a clock: a boundary nobody can land on is a boundary nobody has checked. Two tests land on it — exactly at, and exactly one millisecond past.

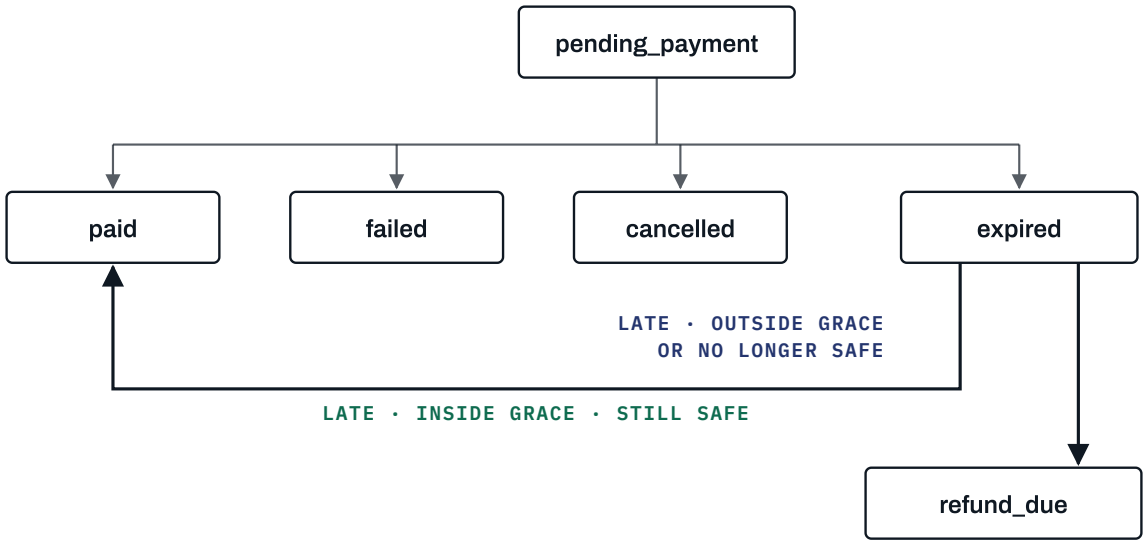
Why 24 hours

Mobile-money confirmations in Niger travel through a carrier and an aggregator before they reach Afrinext. A delay of seconds is ordinary; a delay of hours happens. Refusing those would take money from people who genuinely paid and hand every one of them to support, for no benefit to anybody. Beyond a day the balance tips: the buyer has moved on, may have bought again, and fulfilment becomes a surprise rather than a service.

SETTING	VALUE	HOW IT IS READ
`checkout.ttl_seconds`	1800 · 30 min	Seeded explicitly. The code constant is now a fallback, not the source.
`checkout.late_payment_grace_seconds`	86400 · 24 h	**A ceiling, not a default.** The loader clamps to it, so an operator may narrow the window to zero without asking anyone; widening it is a code change and a review. A malformed value falls back to 24 h — never to zero, never to no limit.

The asymmetry is deliberate and tested: a stored `999999999` still yields 24 hours, and a stored `null`, `"soon"`, `-1` or an object yields 24 hours rather than the zero a coercing read would have produced.

3 The state machine



expired is the ONLY state an order can leave — money can arrive after it. Every other state is terminal.

The two thick arrows are new. **Payments are unchanged**: `initiated → pending → succeeded | failed | cancelled | expired`, each terminal.

Both new transitions are guarded `UPDATE`s naming `status = 'expired'`, and the `CHECK` constraint on `orders.status` was widened to include `refund_due` — a test writes `'refunded'` and requires the database to refuse it.

4 Financial invariants

1 No successful payment is invisible

Every `succeeded` payment leaves its order in `paid` or `refund_due`. Nothing else is a legal resting place, and reconciliation looks for violations.

2 No double fulfilment

One entitlement per buyer per product, by unique index. A late payment for something already owned becomes `refund_due` rather than being absorbed.

3 No double charge swallowed in silence

If a buyer paid twice, the second payment is queued and countable rather than consumed.

4 Fee snapshots are unchanged

A late-accepted order freezes its commission through the *same function* as an on-time one — the two paths were merged into one `fulfil()` precisely so they cannot drift apart.

5 The ledger is untouched

Neither acceptance nor `refund_due` posts a ledger entry. Payment confirmation and financial settlement remain separate, and settlement is still not implemented.

5 Concurrency: the row lock, and the test that needs it

The confirmation transaction takes `SELECT … FOR UPDATE` on the order row **before deciding anything.** A confirmation and the expiry sweeper queue rather than interleave: whichever arrives second sees the state the first committed and takes the branch that state deserves.

> **PROVED DETERMINISTICALLY, NOT BY HOPING TWO PROMISES INTERLEAVE**
>
> A test opens a **second connection**, takes `for update` on the order, and holds it. The confirmation arrives and blocks. The holder then expires the order and commits. Only then does the confirmation get to look — and it sees `expired`, takes the late path, and accepts.
>
> **Without the lock that test fails**, and it fails in the exact shape this gate exists to abolish: the confirmation reads the pre-expiry snapshot, takes the ordinary path, and its guarded `UPDATE` then matches nothing — payment succeeded, order expired, orphaned.

6 Commerce reconciliation

Ledger reconciliation asks whether the books balance. It could not ask whether a succeeded payment left its order somewhere sensible, because it knows nothing about orders — and that gap is how the original defect stayed silent. `reconcileCommerce()` is the missing half:

IMPOSSIBLE STATE	WHY IT MATTERS
`succeeded_payment_orphaned`	Money arrived and left no commercial trace. The original defect.
`paid_order_without_payment`	Fulfilment without money — the mirror image, and worse for the seller.
`multiple_succeeded_payments`	The buyer was charged twice. The partial unique index bounds live attempts, not successful ones over time.
`refund_due_without_payment`	Either the queue is wrong or the payment record is.
`duplicate_entitlement`	Unreachable today; the check is what notices if that unique index is ever dropped.

> **THE GATE PROVES IT CAN STILL SPEAK**
>
> CI asserts the result is empty — but a detector that reports “clean” is worth nothing unless it can report something else, and the suite truncates, so on some runs there would be no orders left and the assertion would pass *vacuously*. That is exactly the failure this gate was opened for.
>
> So on every run the script builds one consistent order-and-payment pair, requires silence, breaks one invariant, **requires the detector to name it**, and puts it back. A run that cannot demonstrate detection fails the build.

7 Test and mutation evidence

```
pnpm lint                          exit 0
pnpm typecheck                     exit 0
pnpm test                          349 passed | 3 skipped (22 files)   was 317
pnpm build                         exit 0
playwright test                    24 passed   production mode, DB from zero
ledger concurrency                 6 passed
reconcile-check                    ledger clean · commerce clean · detector proved
pnpm audit --audit-level high      No known vulnerabilities found
mutation testing                   8 of 8 caught
```

31 new tests in `orders/late-payment.test.ts`, covering all fourteen requested scenarios plus the configuration and state-machine cases, and **2 new tests** in the delivery suite for the two halves of the decision.

> CI
>
> **CI:** run #23 (32414854416) on the code state `4342c06` — `ALL THREE JOBS GREEN`, and run #24 (32415407611) on `9890cdf` — `GREEN AGAIN`. Run #22 on `5d92c8e` was `RED` on the browser job: a pre-existing race in the checkout test, described below and fixed in `4342c06`. `lint · typecheck · test` passed on all three.

DELIBERATE DEFECT	RESULT
The grace-window check is removed	`CAUGHT · 8 TESTS`
The `expired → refund_due` branch is removed	`CAUGHT · 10 TESTS`
The already-owned check is removed	`CAUGHT`
`SELECT … FOR UPDATE` is removed	`CAUGHT`
The grace clamp is removed, so settings can widen the policy	`CAUGHT`
The orphaned-payment reconciliation condition is loosened	`CAUGHT`
The duplicate-payment reconciliation condition is loosened	`CAUGHT`
The entitlement `on conflict do nothing` guard is removed	`CAUGHT`

A SECOND COMMIT: CI FOUND A RACE IN A TEST, NOT IN THE POLICY

The browser job failed on the checkout journey while every local run passed. The test waited for `pay-waiting` — static text inside the form, on screen before the server action has done anything — and then read the payment row, which did not exist yet. Local timing won that race; CI lost it.

The race predates this gate and has nothing to do with the late-payment policy. The order page now shows the charge's real status, which appears only once the action has completed, and the test waits for *that*. It is also better for the buyer: “I have not paid yet” and “the provider has not answered yet” looked identical before.

THE ROW-LOCK MUTATION ESCAPED AT FIRST, AND THE FIX IS THE INTERESTING PART

Removing `FOR UPDATE` changed nothing, because the concurrency test used `Promise.all` and two promises do not reliably interleave two transactions — it created a *possible* race, not a certain one. A test that passes whether or not the lock exists is not testing the lock.

The replacement holds the row lock from a second connection and controls the ordering exactly. It now fails without the lock, every time.

Files changed — 21 across the gate

FILE	WHY
The policy	
core/orders/state.ts	`refund_due` added; `expired` gains two exits and stops being terminal.
core/orders/payment.ts	The row lock, the late decision and its four conditions, the shared `fulfil()`, the injected clock, the decision written to the audit log.
core/orders/checkout.ts	`loadLatePaymentGraceSeconds()` — clamped to the reviewed maximum, malformed values fall back rather than widen.
Reconciliation	
core/ledger/commerce-reconcile.ts	**New.** The five impossible commerce states.
core/scripts/reconcile-check.ts	The CI gate, plus the probe that proves the detector can still report a fault.
core/ledger/index.ts	Export.
Schema and configuration	
db/migrations/0011 + meta/_journal.json	Widened status CHECK, `late_payment_at` and its constraint, the refund-queue index, both settings.
db/schema/commerce.ts	The same, in Drizzle.
core/seed/reference.ts	`checkout.ttl_seconds` and `checkout.late_payment_grace_seconds` seeded explicitly.
Tests	
core/orders/late-payment.test.ts	**New.** 31 tests: all fourteen requested scenarios, the configuration cases, the state machine, and the five reconciliation detections.
core/orders/orders.test.ts	Two tests updated to the approved policy — named below.
core/content/content.test.ts	One updated, one added, for the two halves of the decision.
web/e2e/checkout.spec.ts	Waits for the payment state instead of a static caption — the CI race.
Web and documentation	
web/[locale]/orders/[id]/page.tsx	Shows the charge's real status, and states a refund is owed *and not executed*.
i18n fr.json · en.json	4 keys per locale, worded so nothing claims a refund happened.
docs/architecture/late-payment-policy.md	**New.** The decision, the table, the machine, the invariants, the settings, and what is not built.
AGENTS.md	The rule added to the non-negotiables.
docs/review/… .html · .pdf	This packet.

No new dependency; `pnpm-lock.yaml` is untouched. `ledger/flows.ts`, `ledger/post.ts`, `ledger/accounts.ts`, `money/`, `authz/`, `consent/`, `auth/` and `payments/` have **no diff**.

Three existing tests were updated, and why that is not a weakening

> NAMED HERE RATHER THAN LEFT IN THE DIFF
>
> `orders.test.ts` · **"lets an order leave pending_payment exactly once"** asserted every non-pending state was terminal. `expired` no longer is. It now asserts the new machine *exactly*: expired reaches `paid` and `refund_due` and nothing else, and every other state is still terminal — more specific than before, not less.
>
> `orders.test.ts` · **"refuses a confirmation arriving after the order expired"** asserted the behaviour this gate replaced. It keeps the invariant that matters — a succeeded payment never leaves its order in limbo — and points at the new suite for the edges.
>
> `content.test.ts` · **"grants nothing when the order expired"** now uses a payment past the grace window, so it still proves that a `refund_due` order grants no content. A second test was added for the other half: a payment a minute late *does* deliver the file.

8 Risks, and what is deliberately not built

CONFIRMATION	
No refund execution, through any provider	CONFIRMED
No iPayMoney code, credentials, endpoints, payloads or signatures. The mock remains the only provider, and nothing in this policy is provider-specific	CONFIRMED
No settlement, seller wallet or payout	CONFIRMED
No ledger architecture change. `ledger/flows`, `ledger/post`, `ledger/accounts` and `money/` have no diff; entitlement and refund_due post nothing	CONFIRMED
Webhook, payment and entitlement idempotency preserved and re-tested under the new paths	CONFIRMED
Authorization, consent, OTP and pricing behaviour untouched	CONFIRMED
No new dependency	CONFIRMED

Open risks

ITEM	STATE
Nothing drains the `refund_due` queue. The state is correct, countable and visible to the buyer; the money still has to be returned by a human, out of band, until refund execution exists. **That is the next thing I would build.**	NEXT
No operator screen lists the queue. Finding it today is a SQL query or the reconciliation report. Nobody is alerted.	OPEN
The buyer is not notified. Their order page says a refund is owed and not yet executed; nothing pushes that to them, because no messaging provider is chosen.	BLOCKED ON SMS PROVIDER
Reconciliation is run by CI, not on a schedule. In production it needs a job and an alarm; neither exists.	BEFORE LAUNCH
A late acceptance re-opens a closed order, which is a real change in behaviour a seller may notice as a sale appearing a day later. Deliberate, and the reason the window is a day rather than a week.	ACCEPTED
The grace window is measured from `expires_at`, not from when the buyer left for the provider. An order that expired early for another reason inherits the same window.	ACCEPTED

Is Phase 2 ready for approval? The financial policy gate is closed: the rule is decided, documented, implemented, enforced by guarded transitions and a row lock, and watched by a reconciliation that proves it can still report a fault. What is *not* closed is the consequence of that rule — **a refund queue with no drain.** My recommendation is to approve the gate and treat refund execution as the first item of the next milestone, before settlement, because it is the only open item that can leave a real buyer out of pocket.

> **PROVENANCE, REPEATED — IDENTICAL TO PAGE ONE**
>
> **Branch:** `claude/create-app-repository-1cxsih` · not merged into `main`.
> **Code state under review:** `4342c06` · CI run #23 green.
> **Re-verified at:** `9890cdf` · CI run #24 green.
> **Gate commits:** `5d92c8e` → `4342c06` → `9890cdf` → this reissue.
> **Proof the packet is not stale:** `git diff 4342c06 HEAD -- ':!docs'` returns nothing.
>
> **Final results:** 349 domain tests · 24 browser tests · 8 of 8 mutations caught · ledger and commerce reconciliation clean · dependency audit clean.